



Eutrophication drives functional and beta diversity loss in epiphytic cyanobacteria

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Abstract The structure of epiphytic cyanobacteria and patterns of β - and functional diversity are crucial for understanding how aquatic environments respond to disturbances. This study evaluated how epiphytic cyanobacteria's biomass, adaptive strategy groups, and β - and functional diversity components respond to increasing eutrophication and environmental homogenization in three tropical reservoirs. Hypereutrophic conditions supported greater biomass of competitive species (C-strategists), while stress-tolerant

(S-strategists) were more abundant in oligo-mesotrophic waters. In contrast, ruderal species (R-strategists) were most abundant in the eutrophic system and influenced by rainfall. Total β -diversity and turnover were significantly lower in eutrophic and hypereutrophic systems, correlating with greater environmental homogenization. The oligo-mesotrophic reservoir presented a higher local contribution to β -diversity, highlighting the presence of unique species. Communities in the hypereutrophic system were subsets of species from richer communities, exhibiting higher nestedness. Eutrophication also negatively impacted functional diversity, with eutrophic and hypereutrophic environments showing lower functional evenness, divergence, dispersion, and dissimilarity, and higher functional redundancy due to the loss of unique functions. These results emphasize the detrimental impact of eutrophication on β - and functional diversity of epiphytic cyanobacterial communities, underscoring the need for eutrophication control measures in tropical reservoirs.

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Introduction

Global freshwater biodiversity is facing major declines at an alarming rate (Barnosky et al., 2011;

Albert et al., 2021; Sayer et al., 2025), with severe consequences for ecosystem functioning and services (Cardinale et al., 2012). Man-made reservoirs are particularly subject to anthropogenic, climatic, and hydrological disturbances (Meerhoff et al., 2022; Priya et al., 2023). These disruptions are caused by nutrient enrichment, warming, and reduced light penetration (Mi et al., 2023), further intensified by pronounced seasonal rainfall variability in tropical and semiarid regions (Santos & de Farias, 2017). During extended dry periods, declining water levels in reservoirs increase nutrient concentrations, leading to eutrophication and cyanobacteria blooms (Amorim et al., 2020; Bonilla et al., 2023). Among these nutrients, nitrogen and phosphorus boost algal growth and impact the trophic structure of aquatic systems (Paerl et al., 2016).

Cyanobacteria are widespread in aquatic systems and can occur in planktonic form when living freely in the water column, and in benthic communities when adhered to substrates such as aquatic macrophytes (Reynolds, 2006). Epiphytic cyanobacteria and other periphytic microorganisms are involved in nutrient cycling, primary production, and ecosystem functioning. Macrophytes, in particular, are crucial as substrates for these communities, as their morphological complexity, life forms, and physiological traits support the establishment of diverse autotrophic and heterotrophic assemblages (do Nascimento Filho et al., 2021). Periphytic biofilms on these plants regulate several ecological processes, such as nutrient uptake, pollutant removal, and shading, contributing to ecosystem stability and serving as bioindicators of water quality (Wijewardene et al., 2022). While planktonic cyanobacteria have been extensively studied, epiphytic and benthic cyanobacterial communities remain underexplored (Burford et al., 2020; Wood et al., 2020), especially in tropical regions, such as Northeast Brazil, where only a few studies have examined benthic cyanobacterial communities (e.g., Borges et al., 2015; do Nascimento Filho et al., 2021, 2024).

The C-S-R functional strategy group conceptual model was proposed by Biggs et al. (1998) based on the C-S-R life history strategies of terrestrial plants (Grime, 1977). This approach is useful for examining adaptive strategies within benthic communities. In this model, cyanobacteria and other algae are classified as (i) C-strategists (competitive), with

preferences for eutrophic and stable conditions; (ii) C-S-strategists (competitive and stress-tolerant), adapted to mesotrophic and undisturbed conditions; (iii) S-strategists (stress-tolerant), tolerant to oligotrophic and steady systems; and (iv) R-strategists (ruderal), adapted to heterogeneous habitats with varying nutrient levels (Biggs et al., 1998).

Nutrient levels and disturbance regimes significantly influence the spatial turnover and functional diversity of cyanobacteria in reservoirs. Research on plankton communities indicates that eutrophication increases similarity in cyanobacterial communities, while functional diversity drives community stability (Shen et al., 2024). This phenomenon is further supported by a loss of β -diversity, with communities becoming more similar over time due to anthropogenic impacts and eutrophication (Monchamp et al., 2017). Environmental heterogeneity has been known to increase taxonomic diversity for decades (Ricklefs, 1977; Stein et al., 2014; Agra et al., 2023). Eutrophication, in turn, is one of the main drivers of environmental and biotic homogenization, threatening freshwater biodiversity (Salgado et al., 2018). However, eutrophication can increase taxonomic diversity and decrease the phylogenetic diversity of planktonic cyanobacteria, leading to homogeneous community structures (Zuo et al., 2024). While little is known about benthic communities, the stability of free-living bacteria is greater than that of particle-attached bacteria under hypereutrophic conditions (Xie et al., 2024). The β - and functional diversity of benthic communities are influenced by various environmental and ecological factors, including species dispersal, habitat complexity, and functional traits (Petsch et al., 2017; Dunck et al., 2019).

The alarming rates of freshwater species extinctions in the Anthropocene are primarily driven by eutrophication and water pollution (Sayer et al., 2025). This phenomenon significantly impacts numerous ecosystem services such as food security, underscoring the urgent need to reduce external nutrient inputs into freshwater systems (Jeppesen et al., 2025). Examining how eutrophication affects benthic cyanobacterial communities, along with their β - and functional diversity, is particularly crucial for understanding their resilience and ecological roles, since they are important primary producers, vital food sources for aquatic organisms, and can potentially form toxic mats.

Accordingly, this study investigates the structure of epiphytic cyanobacterial communities on *Pontederia crassipes* Mart. (formerly *Eichhornia crassipes* (Mart.) Solms) across three tropical reservoirs with different trophic states. We analyzed the structure, β - and functional diversity components of these communities, testing three hypotheses: (i) Competitive species (C-strategists) thrive in hypereutrophic reservoirs with stable and nutrient-rich conditions, while stress-tolerant taxa (S-strategists) are abundant in nutrient-poor and heterogeneous oligo-mesotrophic environments; (ii) turnover is the primary driver of β -diversity, which decreases with eutrophication, with a consequent increase in nestedness due to a loss of unique species; and (iii) eutrophication reduces environmental heterogeneity with consequent decreases in functional diversity (i.e., higher functional redundancy), leading to functional homogenization compared to higher dissimilarity in oligo-mesotrophic environments.

Materials and methods

Study area and sampling

Sampling was conducted quarterly in 2022 in three reservoirs in Pernambuco, Brazil: Cursai, Apipucos, and Tapacurá (Table 1; Fig. S1). Two samples were collected during the rainy season (March and June 2022) and two during the dry season (September and

December 2022) in four distinct sampling sites in each reservoir. The reservoirs were selected based on their trophic states and multiple uses. Cursai (oligo-mesotrophic; Amorim & Moura, 2022; Diniz et al., 2023) is inserted in an agricultural landscape and is primarily used for public supply, irrigation, fishing, and recreation. Apipucos (eutrophic; Severiano et al., 2021) is inserted in a densely populated area subject to intense anthropogenic activities (urbanization and industrial waste disposal), and is used for recreation, fishing, and flood control. Despite being in a natural reserve, the land use of Tapacurá's drainage basin (hypereutrophic; Amorim & Moura, 2022) is mainly composed of pasture, agriculture, and urban areas. Tapacurá serves for flood control, water supply, fishing, and irrigation in agriculture. The geographical location and the main land uses of the reservoirs' drainage basins are shown in Fig. S1.

Meteorological variables, such as rainfall (mm) and wind speed (km h^{-1}), were obtained from nearby weather stations (APAC, 2023). The depth and water quality variables were measured in situ, including temperature ($^{\circ}\text{C}$), electrical conductivity ($\mu\text{S cm}^{-1}$), dissolved oxygen (mg L^{-1}), turbidity (NTU), and pH using a HANNA multiparameter probe (HI 9829). Water transparency (m) was estimated using a Secchi disk. A principal component analysis (PCA) was employed to assess environmental heterogeneity using standardized (z-scores) environmental variables with the R package “vegan” (Oksanen et al., 2024). The PC1 scores were standardized to minimum and

Table 1 Geographical location, municipality and its current population, storage volume, trophic state, drainage area, and main land use types in the studied reservoirs

	Cursai	Apipucos	Tapacurá
Latitude	7°52'33.23"S	8°1'12.8"S	8°2'36.2"S
Longitude	35°10'42.2"W	34°55'58.4W	35°11'44.8"W
Municipality	Paudalho	Recife	Vitória de Santo Antão ^e
Population in 2022 (inhabitants) ^a	56,655	1,488,920	134,084
Storage volume (m^3) ^b	13,033,990	278	104,870,609
Trophic state	Oligo-mesotrophic	Eutrophic	Hypereutrophic
Drainage area (km^2) ^c	58.4	3.4	377.6
Forest (%) ^d	2.1	2.9	5.0
Pasture and agriculture (%) ^d	88.0	5.5	80.6
Urban or build-up areas (%) ^d	6.2	89.1	10.9

^a Obtained from <https://censo2022.ibge.gov.br/>; ^b Obtained from <https://www.apac.pe.gov.br/>; ^c Calculated using the plugin GRASS on QGIS; ^d Obtained from <https://brasil.mapbiomas.org/>; ^e Tapacurá is located at the border between Vitória de Santo Antão and São Lourenço da Mata, but Vitória de Santo Antão is located upstream and contributes more to land cover of Tapacurá's drainage basin

maximum values ($x' = [x - \min] / [\max - \min]$) to account for negative and near-zero values. The coefficient of variance of standardized PC1 scores was used as a proxy for environmental heterogeneity within each reservoir and season (according to Agra et al., 2023).

Cyanobacteria biomass and functional strategy group quantification

Epiphytic cyanobacteria were collected from submerged petioles of *P. crassipes*, a macrophyte common in all reservoirs. This macrophyte is a perennial free-floating aquatic plant, with dense axial roots and floating petioles containing aeriferous parenchymal tissues (Gopal & Chamanlal, 1991). Sampling was conducted by scraping a standardized area of 9 cm² from 2–3 submerged petioles per plant using a brush with soft bristles. The samples were preserved with 4% formalin for qualitative analyses and with 1% Lugol's solution for quantitative analyses.

Species identification was performed using optical microscopy, with assistance from specialized taxonomic literature (e.g., Komárek & Anagnostidis, 1986, 2007; Komárek, 2013), and taxonomical updates based on the AlgaeBase database (<https://www.algaebase.org/>). The species were quantified using an inverted microscope (BIOVAL—XDS-1D) following Utermöhl (1958), with sedimentation time based on Lund et al. (1958). Cyanobacteria biovolume was calculated according to Hillebrand et al. (1999) and then converted to biomass (mg cm⁻²). Species were classified into C-S-R functional strategy groups based on their morphology, ecology, and the environments they inhabit (Biggs et al., 1998). S-strategists are characterized by prostrate filaments surrounded by mucilage, capable of nitrogen fixation, and composed of small cells. C-S-strategists include heterotrichous or prostrate filamentous and unicellular species, with small to medium cells. C-strategists are upright filaments, with or without mucilage, and consist of medium to large cells. R-strategists are unicellular or colonial species with mucilage, comprised of small cells, reproduce through propagule production, and are resistant to disturbance (Biggs et al., 1998). Functional traits, including size classes (ordinal: <30, 30–80, or >80 µm), growth form (categorical: unicellular,

colonial, or filamentous), substrate attachment capacity (categorical: firmly or loosely attached; Sládecková & Sládeček, 1977), functional strategy group (categorical: C-, S-, R-, and C-S-strategists; Biggs et al., 1998), presence of mucilage sheaths, and N₂ fixation (binary), were evaluated.

Total biomass and each strategy group were compared with a two-way analysis of variance (ANOVA), testing the isolated and interactive effects of reservoirs and seasons. The normality and homogeneity of the residuals were assessed using the Kolmogorov–Smirnov and Bartlett's tests, respectively. When those assumptions were not met, a nonparametric aligned rank transformation analysis of variance (ART-ANOVA) model was applied using the R package “ARTool” (Wobbrock et al., 2011). A redundancy analysis (RDA) was conducted to assess the influence of environmental variables on the functional strategy group structure. The biomass data were transformed (Hellinger) and the environmental variables were standardized (z-scores). Autocorrelated variables with VIF values greater than 5 were removed. The significance of the RDA model, variables, and axes was tested via permutation-based ANOVA (999 permutations). The RDA was performed using the R package “vegan.”

β- and functional diversity indices

Total β-diversity and its components (turnover and nestedness) were determined following Baselga's method (Baselga, 2010). The Sørensen index was calculated to assess total β-diversity (β_{sor}), which was decomposed into turnover (Simpson pairwise dissimilarity, β_{sim}) and nestedness-resultant dissimilarity (β_{nes}) components, using the R package “betapart” (Baselga & Orme, 2012). Principal coordinates analyses (PCoA) of each dissimilarity matrix (total, turnover, nestedness) were employed to visualize community differences across reservoirs and seasons. A permutational multivariate analysis of variance (PERMANOVA) tested the differences between β-diversity and its components across reservoirs, seasons, and their interaction. A Euclidean dissimilarity matrix was computed for the environmental variables, and each β-diversity component was correlated with environmental dissimilarity via Mantel tests (with 999 permutations).

The PCoA, PERMANOVA, and Mantel tests were performed in the R package “vegan.” The correlation between the environmental factors was assessed using Spearman’s correlation. The local contributions to β -diversity (LCBD) and species contributions to β -diversity (SCBD) were calculated to identify key sites and taxa contributing to compositional variability according to Legendre & De Cáceres (2013).

The functional diversity of benthic cyanobacterial communities was assessed using trait and biomass matrices in the package “FD” (Laliberté & Legendre, 2010; Laliberté et al., 2014). Trait data were transformed using the Gower’s dissimilarity (Podani, 1999), and autocorrelated traits were removed (i.e., growth form and N_2 fixation). The indices included Functional Richness (FRic), Functional Evenness (FEve), Functional Dispersion (FDis), and Functional Dissimilarity (Rao’s quadratic entropy, RaoQ). Functional redundancy was calculated as the ratio between trait dissimilarity (RaoQ) and taxonomic diversity (D: Simpson’s index) ($FRed - 1 \text{ RaoQ/D}$; Ricotta et al., 2016). Rao’s quadratic entropy is the best index to represent functional diversity because it involves species abundances and multiple traits, often surpassing other indices (Botta-Dukát, 2005). These metrics, in addition to the LCBD results, were compared with a two-way ANOVA or ART-ANOVA, testing

the isolated and interactive effects of reservoirs and seasons, as previously described.

Results

Environmental heterogeneity

The oligo-mesotrophic reservoir presented higher pH and depth, lower electrical conductivity, water transparency, and wind speed than the eutrophic and hypereutrophic reservoirs. Dissolved oxygen and temperature remained similar across reservoirs and seasons, while rainfall was higher in the rainy season. The PCA biplot and coefficient of variance values revealed that the oligo-mesotrophic reservoir exhibited up to eight times greater environmental heterogeneity than the eutrophic and hypereutrophic reservoirs, mainly in the rainy season (Fig. 1).

Structure and drivers of benthic cyanobacterial communities

The biomass of benthic cyanobacteria and S- and C-S-strategists varied across reservoirs and seasons, whereas C- and R-strategists differed only across reservoirs. Higher total biomass and biomass of C-S-strategists were observed in the oligo-mesotrophic (Cursai) and the hypereutrophic (Tapacurá) reservoirs during the dry season. S-strategists exhibited higher

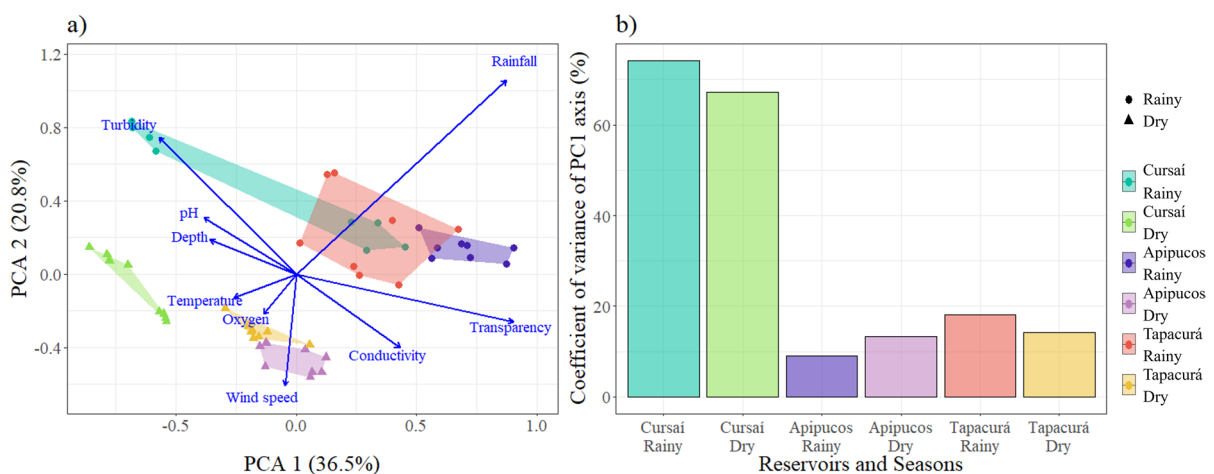


Fig. 1 Principal component analysis (PCA) (a) and coefficient of variance of PC1 axis (b) to represent environmental heterogeneity in three tropical reservoirs (Cursai: oligo-mesotrophic;

Apipucos: eutrophic; Tapacurá: hypereutrophic) and two seasons (rainy and dry)

biomasses in the oligo-mesotrophic reservoir during the rainy season, followed by the hypereutrophic and eutrophic (Apipucos) reservoirs. C-strategists were more abundant in the hypereutrophic, while R-strategists were most prevalent in the eutrophic reservoir (Fig. 2; Table S1).

A redundancy analysis (RDA, $F=5.119$, $P<0.001$) explained 54.8% of the total variance in the functional strategy group biomass, with the first two axes accounting for 93% of the constrained variance. S-strategists were linked with higher pH, temperature, turbidity, and depth, particularly in the oligo-mesotrophic reservoir during the rainy season. C-S-strategists correlated with wind speed and were associated with the hypereutrophic reservoir and the oligo-mesotrophic during the dry season. R-strategists positively correlated with electrical conductivity and rainfall in the eutrophic reservoir. Lastly, C-strategists were most abundant in the hypereutrophic reservoir, though their distribution was not influenced by environmental variables (Fig. 3).

β -diversity components

Total β -diversity was mainly driven by turnover, which was significantly distinct in the oligo-mesotrophic reservoir compared to the others, while nestedness did not vary across reservoirs. The eutrophic and hypereutrophic reservoirs showed less variation in total β -diversity and turnover, whereas the oligo-mesotrophic reservoir exhibited a wider variability, which also differed from the others (Fig. 4a–c; Table S2). Temporal total β -diversity did not differ across reservoirs. However, temporal turnover was significantly higher, and temporal nestedness was substantially lower in the oligo-mesotrophic reservoir compared to the hypereutrophic reservoir. The eutrophic reservoir displayed intermediate values for all components of β -diversity (Fig. 4d; Table S3). Spatial total β -diversity and its components did not differ across months, but total β -diversity and turnover were significantly higher in the rainy season (Fig. 4e; Table S3). Total β -diversity ($r=0.315$, $P<0.001$) and turnover ($r=0.303$, $P<0.001$) matrices were significantly correlated with the environmental dissimilarity matrix, but nestedness was not ($r=0.042$, $P=0.786$). Variables such as turbidity, followed by transparency,

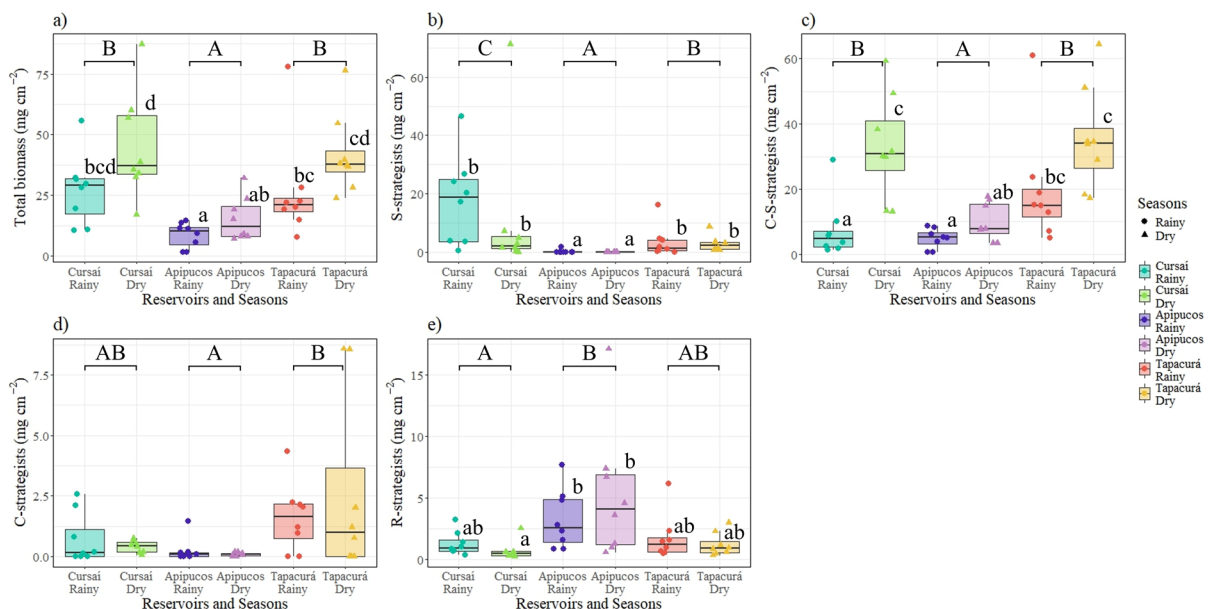
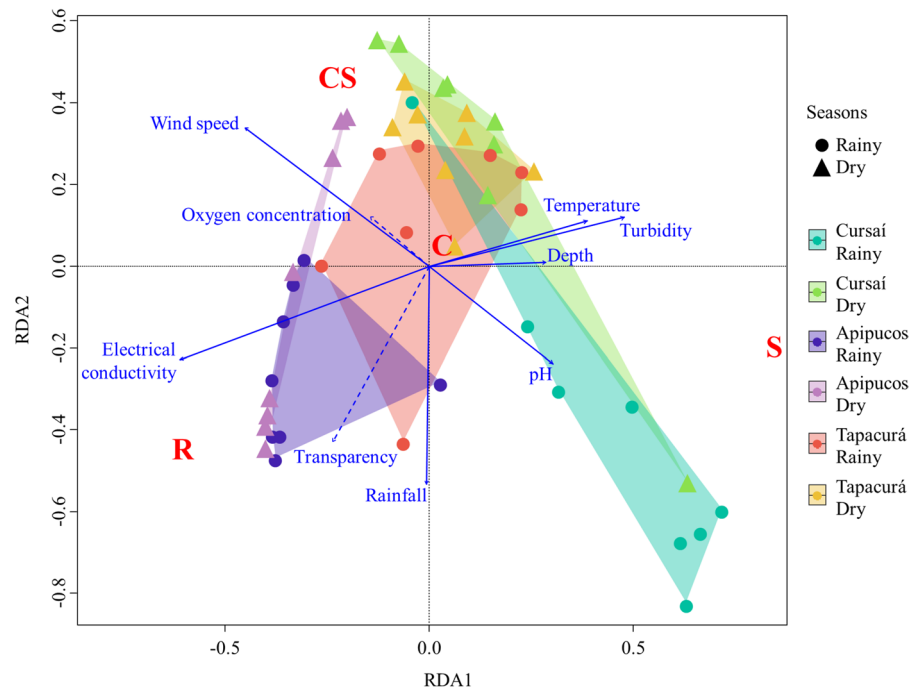


Fig. 2 Total epiphytic cyanobacteria biomass (a), S-strategists (b), C-S-strategists (c), C-strategists (d), and R-strategists (e) across three tropical reservoirs (Cursai: oligo-mesotrophic; Apipucos: eutrophic; Tapacurá: hypereutrophic) and two sea-

sons (rainy and dry). Different lowercase letters represent significant differences between treatments and uppercase letters between reservoirs

Fig. 3 Redundancy analysis (RDA) showing the correlations between environmental variables and functional strategy groups of epiphytic cyanobacterial communities in three tropical reservoirs (Cursai: oligo-mesotrophic; Apipucos: eutrophic; Tapacurá: hypereutrophic) and two seasons (rainy and dry). Significant variables ($P < 0.05$) are represented with solid lines



depth, pH, wind speed, electrical conductivity, and rainfall influenced total β -diversity and turnover, while nestedness was affected only by wind speed. The environmental variables showed few and weak correlations (Fig. S2).

The local contribution to β -diversity (LCBD) significantly differed between reservoirs and seasons. The oligo-mesotrophic reservoir contributed more to the β -diversity than the eutrophic and hypereutrophic reservoirs, with higher values in the rainy season (Fig. 5a; Table S3). For the species contribution to β -diversity (SCBD), the most important species were R-strategists (*Aphanocapsa incerta* (Lemmermann) G.Cronberg & Komárek, *Coelosphaerium kuetzingianum* Nägeli, *Chroococcus minutus* (Kützinger) Nägeli, and *Merismopedia tenuissima* Lemmermann), followed by C-S-strategists (*Pseudanabaena galeata* Böcher, *Pseudanabaena mucicola* (Naumann & Huber-Pestalozzi) Schwabe, and *Geitleribactron periphyticum* Komárek), C-strategists (*Planktothrix agardhii* (Gomont) Anagnostidis & Komárek), and S-strategists (*Calothrix fusca* Bornet & Flahault). SCBD values positively correlated with the frequency of occurrence of the species, except for the most frequent species (Fig. 5b; Fig. S3; Table S4).

Functional diversity

All functional diversity metrics varied across reservoirs, but only functional dispersion and dissimilarity differed between seasons. Functional richness was lowest in the eutrophic reservoir, while functional evenness was highest in the oligo-mesotrophic, followed by the hypereutrophic, and eutrophic systems. Functional divergence, dispersion, and dissimilarity decreased linearly with increasing eutrophication, with the highest values observed in the oligo-mesotrophic reservoir. Notably, functional divergence differed significantly across all reservoirs. Functional dispersion was greater in the oligo-mesotrophic reservoir compared to the hypereutrophic, while functional dissimilarity in the oligo-mesotrophic system surpassed values in both eutrophic and hypereutrophic reservoirs. Higher values of functional dispersion and dissimilarity were recorded in the rainy season. Lastly, functional redundancy was significantly greater in the eutrophic and hypereutrophic systems than in the oligo-mesotrophic reservoir (Fig. 6; Table S5).

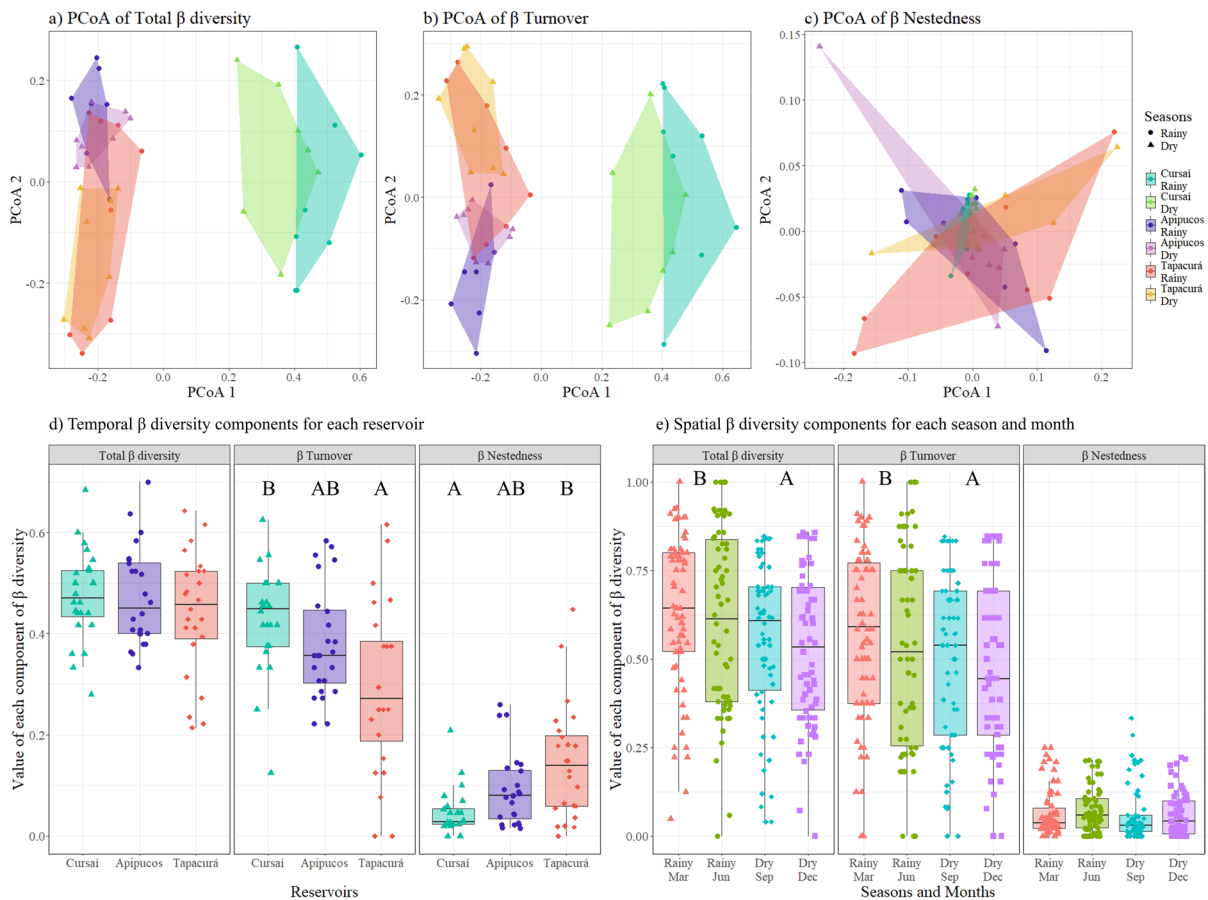


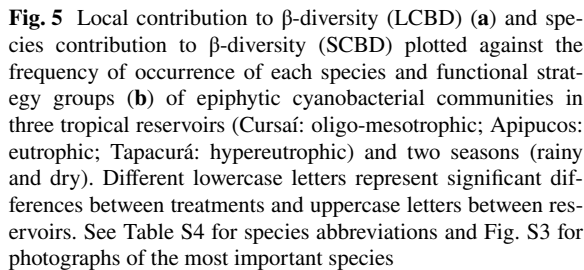
Fig. 4 Principal coordinates analysis (PCoA) comparing total β -diversity (a), turnover (b), and nestedness (c) components of epiphytic cyanobacterial communities in three tropical reservoirs (Cursai: oligo-mesotrophic; Apipucos: eutrophic; Tapacurá: hypereutrophic) and two seasons (rainy and dry). Tempo-

ral (d) and spatial (e) β -diversity components for each reservoir and season, respectively. Different uppercase letters represent significant differences between reservoirs (d) and seasons (e). The legend on the right is for the figures (a–c)

Discussion

Our results demonstrated that the biomass of epiphytic cyanobacteria and functional strategy groups, as well as β - and functional diversity metrics, were significantly influenced by eutrophication. Overall, the homogeneous hypereutrophic reservoir showed a higher contribution of C-strategists; S-strategists were more abundant under oligo-mesotrophic and heterogeneous conditions; whereas R-strategists were more abundant in the eutrophic system. β -diversity was mainly represented by the turnover component, which was significantly higher in the oligo-mesotrophic reservoir, and temporal nestedness was considerably lower. The eutrophic

and hypereutrophic reservoirs displayed lower functional evenness, divergence, dispersion, and dissimilarity (i.e., lower functional diversity) but higher functional redundancy (i.e., higher overlap of functions) (Fig. 7). Despite the positive influence of eutrophication and cyanobacterial dominance on alpha diversity and productivity in plankton communities observed in some studies (e.g., Liu et al., 2019; Amorim & Moura, 2021; Zuo et al., 2024), our study shows that eutrophication has negative effects on β - and functional diversity of benthic cyanobacteria.



Benthic communities are more susceptible to environmental variations, such as nutrient input (Ferragut & de Campos Bicudo, 2010; Dunck et al., 2019; Lanza et al., 2024). The distribution of benthic cyanobacteria is strongly correlated with eutrophication gradients (Loza et al., 2013). Seasonal rainfall patterns also influence nutrient dynamics. For instance, intense rainfall events are expected to increase nutrient concentrations in lakes and reservoirs in agricultural landscapes (Sinha et al., 2017). However, lakes and reservoirs in dry

The first hypothesis was confirmed: Competitive (C-strategist) epiphytic cyanobacteria presented higher biomass in the environmentally homogeneous hypereutrophic reservoir, whereas stress-tolerant (S-strategists) were more abundant in the oligo-mesotrophic and heterogeneous reservoir. Eutrophication and other environmental factors, like water flow, electrical conductivity, turbidity, and temperature, are major drivers of benthic biofilm development (Zill et al., 2024). Anthropogenic activities around eutrophic reservoirs further contribute to environmental homogenization, leading to more similar communities across space and time, a loss of unique species, and the establishment of generalist, fast-growing species (C-strategists) (Biggs et al., 1998; Cook et al., 2018).

The C-S-R model is an effective framework for predicting the response of benthic algae to eutrophication gradients (Ferragut & de Campos Bicudo, 2010). C-strategists, which were most abundant in the hypereutrophic reservoir, thrive in nutrient-rich and stable eutrophic environments (Biggs et al., 1998). They were also predominant in the dry season, when nutrient concentrations are typically higher. In contrast, S-strategists, prevalent in the oligo-mesotrophic system mainly in the rainy season, are better adapted to nutrient-poor waters, where they can tolerate nutrient limitations (e.g., through nitrogen fixation) (Biggs et al., 1998). The shift from S-strategists in the rainy season to C-S-strategists in the dry period within the oligo-mesotrophic reservoir likely reflects a response to increased nutrient availability, as C-S-strategists prefer moderately eutrophic conditions. This group was also dominant in the hypereutrophic reservoir, a pattern similarly observed in the eutrophic Okeechobee Lake (Carrick & Steinman, 2001). R-strategists are specialists and exhibit faster growth in disturbed environments with moderate-to-high nutrient supplies, being the first to colonize (Carrick

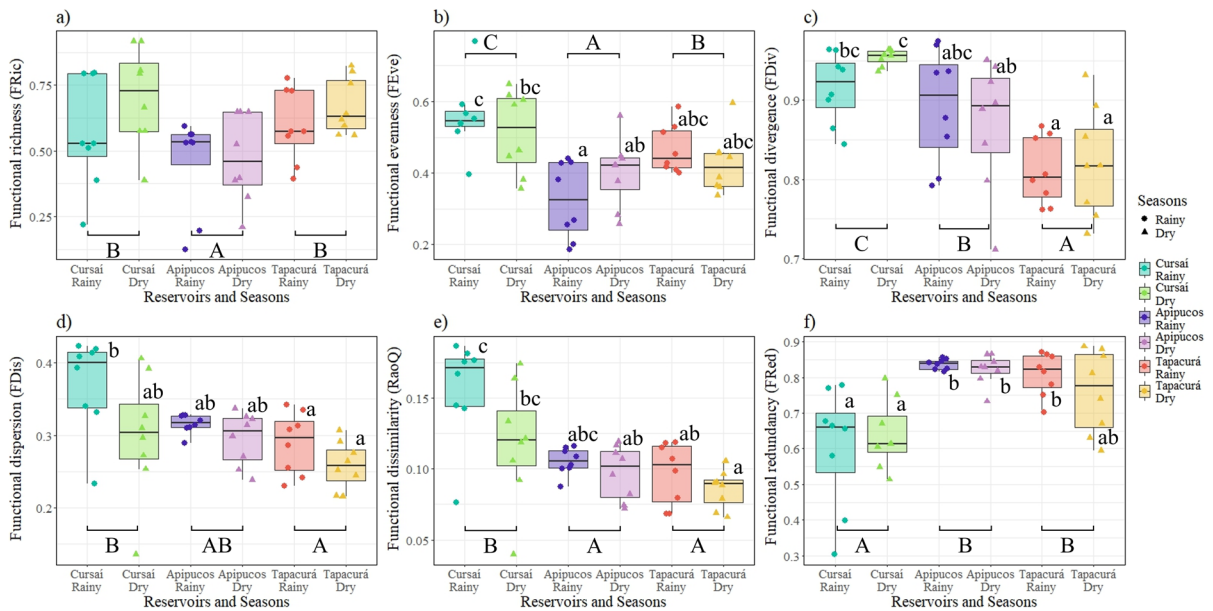


Fig. 6 Functional diversity metrics of epiphytic cyanobacterial communities in three tropical reservoirs (Cursai: oligo-mesotrophic; Apipucos: eutrophic; Tapacurá: hypereutrophic) and two seasons (rainy and dry). **a** Functional richness; **b**

functional evenness; **c** functional divergence; **d** functional dispersion; **e** functional dissimilarity; **f** functional redundancy. Different lowercase letters represent significant differences between treatments and uppercase letters between reservoirs

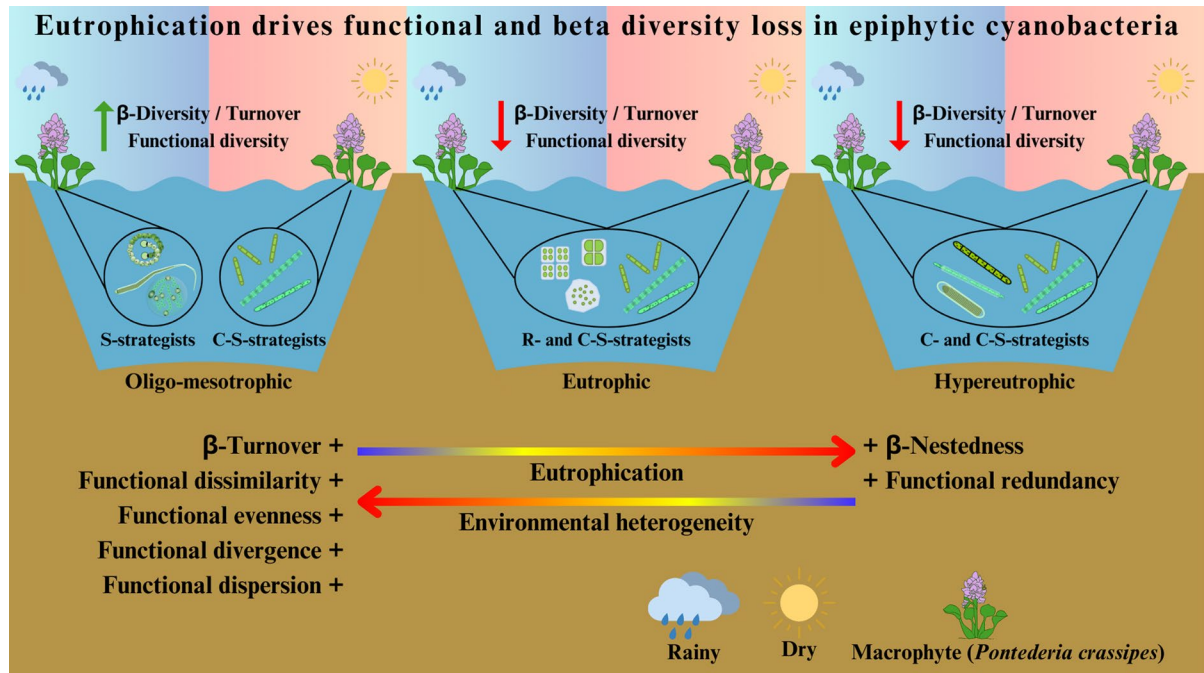


Fig. 7 Summary of the main results found in this study, highlighting the loss of functional and beta diversity in epiphytic cyanobacteria with increasing eutrophication and environmental homogenization

& Steinman, 2001). In our study, this group exhibited higher biomass in the eutrophic reservoir, with rain-fall acting as a disturbance factor (e.g., increasing turbulence) that influenced the epiphytic cyanobacteria community.

Reduced β -diversity and turnover in response to eutrophication

The second hypothesis was also confirmed: β -diversity of epiphytic cyanobacteria was mainly driven by the turnover component and decreased with eutrophication, resulting in an increase in the nestedness component. This result was also observed in phytoplankton communities, where lakes with lower nutrient concentrations host unique communities and higher β -diversity (Maloufi et al., 2016). Local environmental conditions and environmental heterogeneity also promote higher species replacement (i.e., turnover) rather than differences in species richness (i.e., nestedness) (Maloufi et al., 2016). The reduction in β -diversity due to eutrophication occurs because of the dominance of generalist species, promoting biotic homogenization and the development of functionally similar groups. This was observed not only in benthic algal communities (e.g., our study; Passy & Blanchet, 2007; Dunck et al., 2019; Zorzal-Almeida et al., 2021), but also in bacteria (Zuo et al., 2024), phytoplankton, zooplankton (Maloufi et al., 2016; Gianuca et al., 2017; Li et al., 2022; Zuo et al., 2024), aquatic macrophytes (Salgado et al., 2018), and macroinvertebrates (Cook et al., 2018). Since biodiversity potentially enhances ecosystem functioning and stability (Cardinale et al., 2012), β -diversity loss and resulting biotic homogenization will negatively impact ecosystems' resilience to disturbances.

Environmental heterogeneity is one of the main predictors of metacommunity dynamics (Leibold & Norberg, 2004), as supported by the theory of island biogeography (MacArthur & Wilson, 1967). Heterogeneous habitats harbor greater diversity due to the higher variety and availability of resources (St. Pierre & Kovalenko, 2014). In this study, the reduced β -diversity and turnover component in eutrophic and hypereutrophic reservoirs suggest taxonomic homogenization across communities (i.e., higher similarity), enhanced by lower environmental heterogeneity in these ecosystems. Under eutrophic conditions, cyanobacterial communities tend to

become more homogeneous, with decreased species turnover and total β -diversity (Monchamp et al., 2017; Zuo et al., 2024). In contrast, the higher β -diversity and turnover observed in the oligomesotrophic reservoir (i.e., higher environmental heterogeneity) result from the establishment of unique pioneer species, which hinders the development of generalist groups. This phenomenon can be explained by priority effects, where early colonizers prevent the establishment of slow-growing species, leading to distinct communities with unique compositions (Petsch et al., 2017).

The higher nestedness in the hypereutrophic reservoir, which corresponds to low environmental heterogeneity, suggests that eutrophication, in addition to decreasing total β -diversity and species replacement, increases the similarities between communities. These communities become subsets of richer communities, a pattern also observed in benthic diatom communities of eutrophic coastal streams (Costa et al., 2022). Periphytic algae communities (attached to artificial substrates or macrophytes) on simple substrates also display a nested pattern, representing subgroups of species from more complex substrates (e.g., Schneck et al., 2011; do Nascimento Filho et al., 2024). These findings reinforce the negative correlation between habitat complexity (environmental heterogeneity) and community nestedness, as observed in our study.

Loss of functional diversity and increased functional redundancy due to eutrophication

Functional evenness, divergence, dispersion, and dissimilarity decreased with the eutrophication gradient, confirming the third hypothesis. The eutrophic and hypereutrophic systems also showed higher functional redundancy, indicating an overlap in functional traits. This functional homogenization is a result of the establishment of generalist species with similar functions, which prevents the colonization of sensitive species with distinct traits (Lengyel et al., 2023). Similar results were observed by Dunck et al. (2019), who reported lower periphyton functional dissimilarity in eutrophic wetlands. Eutrophication reduces the availability of niches and promotes functional homogenization through trait convergence (Miranda et al., 2024). Therefore, maintaining functional diversity

is important to enhance ecosystem functioning and resilience to eutrophication (Cheng et al., 2023).

Despite presenting higher functional redundancy, the hypereutrophic reservoirs did not show lower functional richness than the oligo-mesotrophic system, indicating that the overlaps in functional traits efficiently fulfill unoccupied niches. Functionally homogeneous communities can still present higher species diversity when there is an overlap in species functions (Mouillot et al., 2014). Higher functional redundancy can also enhance community stability, as these communities become resilient to species loss since more species perform similar ecosystem functions (Biggs et al., 2020), as observed in the eutrophic and hypereutrophic reservoirs in our study. In this scenario, the extinction of a few species would not affect ecosystem functions, as other species with similar functional traits would compensate for the biodiversity loss (Fonseca & Ganade, 2001). However, the oligo-mesotrophic reservoir presented the highest functional diversity and the lowest functional redundancy, highlighting that the species in this environment perform unique functions. Their extinction would disrupt ecosystem stability. Thus, conservation efforts should focus on keeping oligo-mesotrophic systems at low trophic states to enhance β - and functional diversity, as they are more vulnerable to local extinctions.

Conclusions

Eutrophication and environmental homogenization were identified as major drivers of epiphytic cyanobacteria structure, β -diversity, and functional diversity. Hypereutrophic conditions favored competitive generalist species (C-strategists), whereas the oligo-mesotrophic reservoir exhibited higher environmental heterogeneity, promoting stress-tolerant species (S-strategists). These succession patterns increased β -diversity, turnover, and functional diversity in the oligo-mesotrophic system (i.e., unique species and traits) while enhancing nestedness and functional redundancy in the eutrophic and hypereutrophic reservoirs (i.e., subsets of richer communities with overlapping functions). Understanding both β - and functional diversities is crucial for conserving biodiversity and

planning restoration practices to maintain ecosystem stability and functioning (Cadotte et al., 2011; Socolar et al., 2016). Thus, our findings highlight the need to control eutrophication in tropical reservoirs to preserve greater β - and functional diversity, thereby enhancing ecosystem resilience to disturbances.

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Author contributions Fátima S. Silva contributed to conceptualization, methodology, validation, data curation, formal analysis, writing—original draft, and writing—review & editing. Ariadne N. Moura involved in conceptualization, supervision, methodology, and writing—review & editing. Cihelio A. Amorim helped in conceptualization, methodology, validation, data curation, writing—original draft, and writing—review & editing. All authors approved the submission.

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Data availability Additional data may be obtained from the corresponding author (alvescihelio@gmail.com, cihelio.amorim@wcl.ac.at) upon reasonable request.

Declarations

Conflicts of interest Declaration of conflicts of interest: none.

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